

COSTS

Equipment:

- **CZT detector and front-end electronics:** custom multichannel detector system for fluorescence measurements in the 20–50 keV range.
- **Digitizer and DAQ:** multichannel waveform-acquisition system for detector readout, synchronization, and pulse-shape analysis.
- **Motorized positioning stages:** two rotational and one translational stages for the sample holder and CZT detector, for automated switching between fluorescence, $I_f(E)$, and incident-beam, $I_0(E)$, configurations.
- **Cooled X-ray tube and power supply:** High-power W-anode X-ray tube enabling measurements above 20 keV during the second year.
- **Bent germanium crystal optics:** custom curved Ge crystals for monochromatization, energy scanning, and focusing of the incident X-ray beam.

Consumables:

- **Ag reference samples:** metallic Ag foils and oxidized Ag powder standards for calibration, energy-scale verification, and XANES validation.
- **Additional reference targets:** selected materials containing elements with absorption edges above 20 keV, such as Mo, Pd and Sn, for assessing the broader operating range of the instrument.

Services:

- **Custom mechanical design and fabrication:** development of dedicated supports and interfaces for integrating the sample holder, CZT detector, positioning stages, and associated components into the spectrometer.

Computing:

- **CPU and data storage:** computing resources for the SHADOW4–GEANT4 simulations and detector-response studies performed in WP1. The request corresponds to approximately 250,000 core-hours, calculated at a rate of €4.11 per 1,000 core-hours, together with the storage required for simulated events, waveforms, and calibration datasets.

Travels:

- **Conference participation:** travel for the dissemination of project results through an oral presentation. A possible target conference is the European Conference on X-Ray Spectrometry, EXRS 2028, expected to be held in Paris in June 2028.

In-kind contributions:

- PRISM will benefit from existing VOXES infrastructure and equipment during the development and preliminary integration phases. The estimated in-kind contribution includes an X-ray tube available during the first project year (€10k), the shielded experimental bunker (€5k), existing motorized positioning systems (€15k), and linear translation axes with associated control hardware (€15k) for the crystal. The total estimated in-kind contribution is therefore €45k.

Year	Category	Item	Purpose/WP	Cost [€]
2027	Equipment	CZT Detector and Front-End	Fluorescence Detection - WP2	€30.000
2027	Equipment	Digitizer and DAQ	Signal Acquisition - WP2	€23.000
2027	Equipment	Motorized Stages and Controllers	Automated Positioning - WP3	€10.000
2027	Consumables	Ag Reference Targets	Calibration Tests - WP2/WP4	€2.000
2027	Services	Custom Mechanical Supports	Instrument Integration - WP3	€2.000
2027	Computing	Data Cloud CPU and Storage	End-to-End Simulations - WP1	€2.000
Total 2027				€69.000
2028	Equipment	Cooled X-ray Tube	Excitation Source - WP3	€10.000
2028	Equipment	X-ray Tube Power Supply	Source Operation - WP3	€7.000
2028	Travel	Oral Conference Travel	Results Dissemination - WP4	€4.000
2028	Equipment	Bent Crystal Optics	Energy Selection - WP3	€20.000
2028	Consumables	Additional Reference Targets	Multi-Edge Validation - WP4	€2.000
Total 2028				€43.000
Total				€112.000

TABLE I. Detailed financial request for the two-year PRISM project, broken down by year, cost category, item, and related Work Package.